

Accelerating container network with channel based IO

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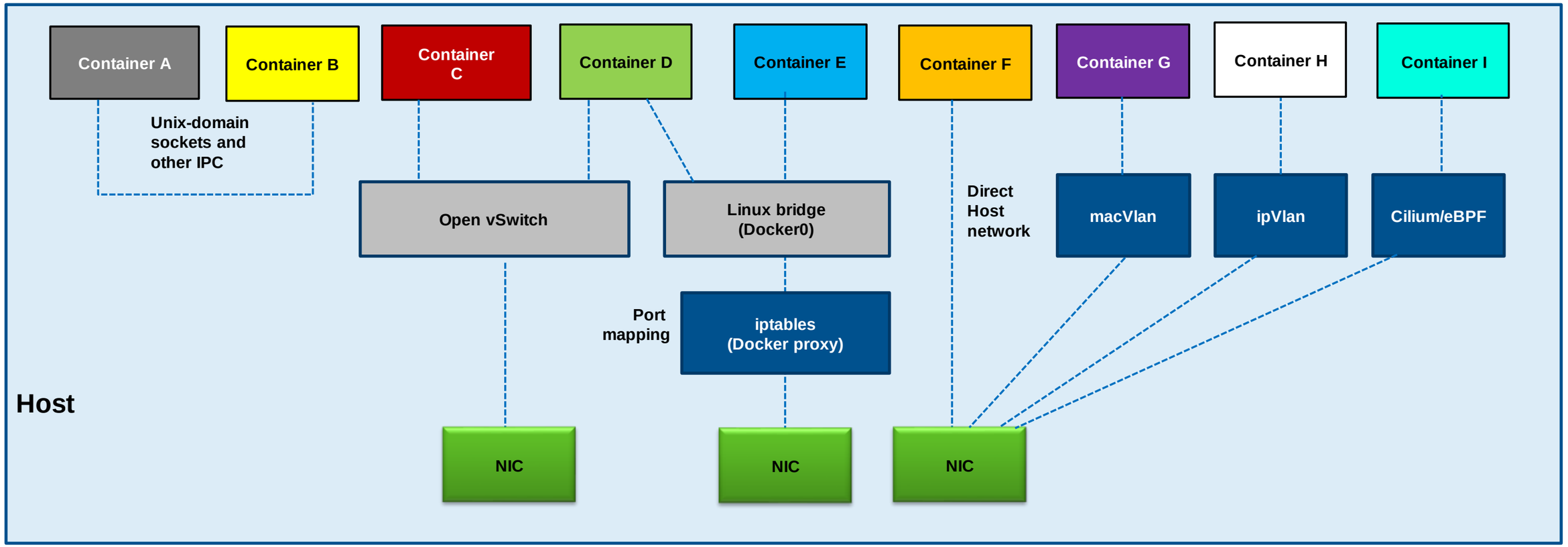
Netdev 1.2 Tokyo Oct 2016

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Multiple options for Container networks

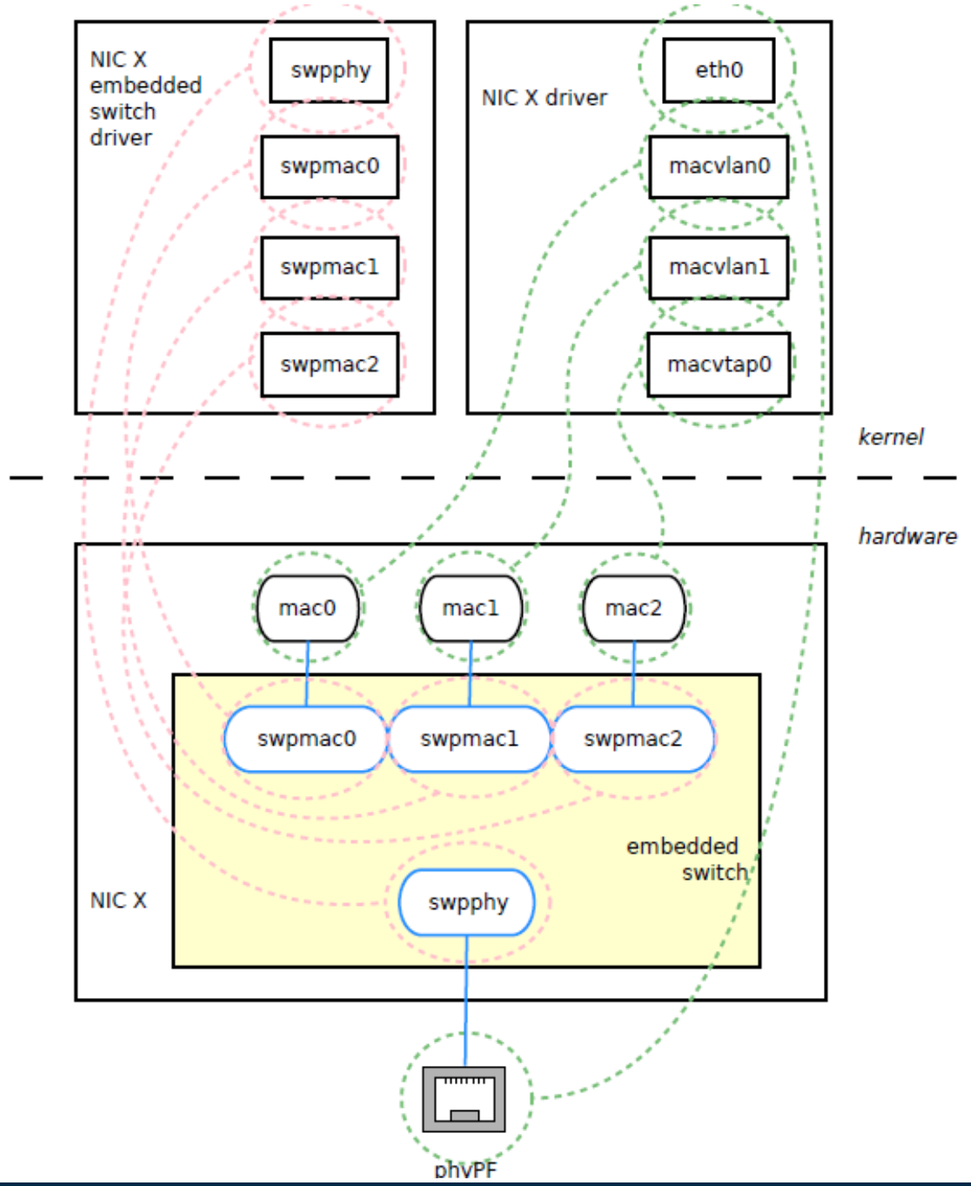
- Managed switch : Linux bridge, openVswitch, Cilium/eBPF
- Unmanaged switch : macVlan ipVlan



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Switch management Module

- in order to config the vPort of the switch we need to have a representor port from the switch side (swpmac)
- The switch vPort is a vendor driver same as VF representor
- The Nic driver is a vendor driver same as VF driver.
- FDB, ACL, flow rules are done by TC

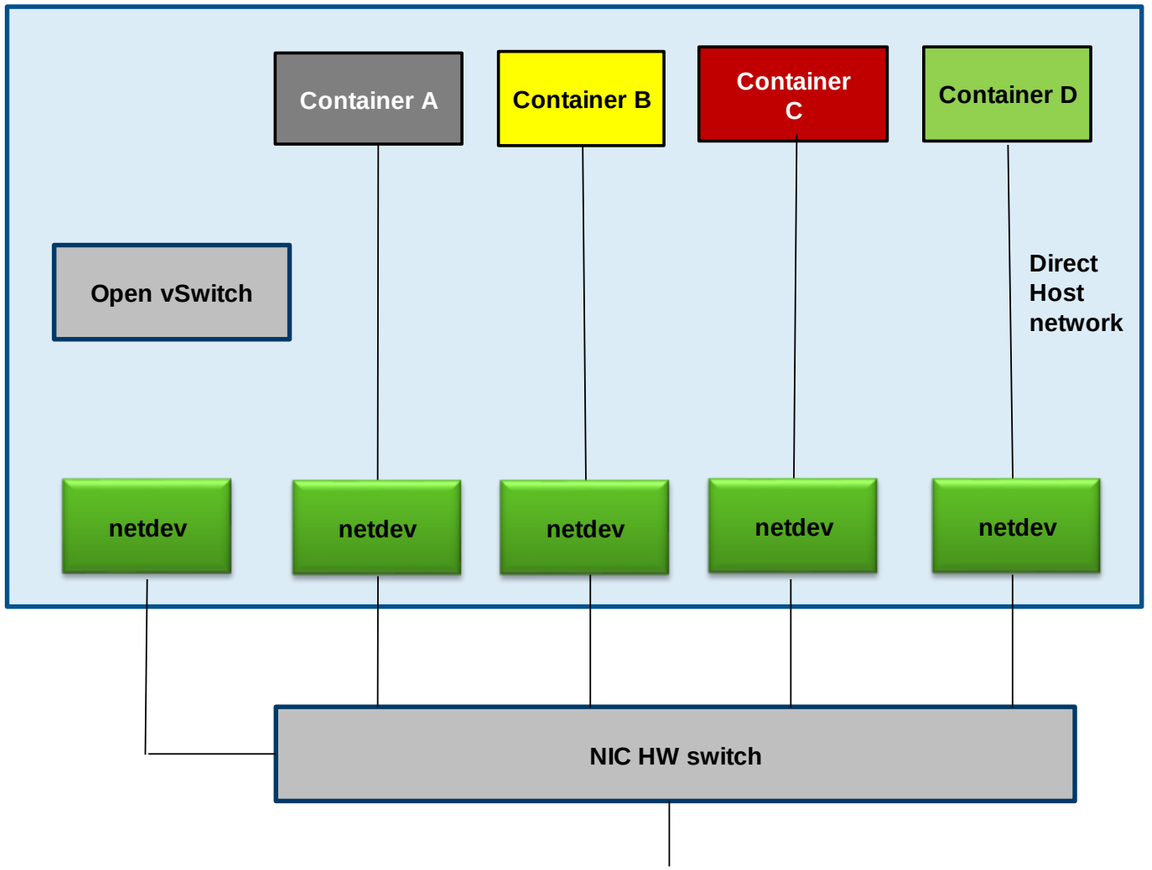


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Roadmap

- Assign HW rings per container
  - Full support of stateless offloads (csum, TSO, LRO)
  - Dedicated RSS indirection table with rings on for container affinities-ed cores
  - QoS – Rate limit and BW reservation per container
  - ACLs – L2-4 based and L7 application specific rules
  - Accelerate Virtual Switch FDB using per container queue
  - HW Encap/Decap for tunneled traffic
  - All the above for tunnel traffic as well (VXLAN, Geneve, GRE)
  - All of the above for containers in SRIOV VMs (nested virtualization)



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How do you see it ?



Thank You

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